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CLOSED-LOOP DUAL-REFERENCE GUIDANCE FOR ACCENT CONTROL IN CROSS-LINGUAL VOICE CLONING

*Yuanzhong Zheng, Wei Huang, Ming Ma, Kailun Jin, Youhai Jiang**

VoxNexus, Singapore

ABSTRACT

Zero-shot cross-lingual voice cloning often carries the reference speaker's accent into the target language. We propose closed-loop dual-reference guidance, which pairs the original reference with a reusable native anchor constructed by voice conversion (VC). The references initialize separate language-model (LM) states that guide a shared acoustic trajectory; each completed patch then updates both states, keeping their generated histories aligned with the emitted speech. Anchor construction runs once per speaker, and online synthesis uses a frozen TTS model without per-utterance VC. On 20 held-out French speakers, the fixed setting $\lambda^* = 0.33$ reduces residual French LID mass by 73% versus direct cloning, at an ECAPA similarity cost of only 0.042; relative to anchor-only cloning, it recovers 0.155. The method achieves the lowest mean WER (0.009) and highest mean GOP (4.750) among the held-out test arms. In blinded preference tests, native-English listeners favor it over direct cloning on pronunciation naturalness, and over both anchor-only cloning and a TTS→VC cascade on speaker identity. Online RTF is approximately 26% lower than the cascade's (0.59 vs. 0.80), and the same setting reduces Spanish LID mass by 56% without re-tuning.

***Index Terms*—** speech synthesis, accent conversion, classifier-free guidance, voice cloning, flow matching

1. INTRODUCTION

Cross-lingual zero-shot text-to-speech (TTS) synthesizes a target language from a short reference in another language. The reference carries both vocal identity and pronunciation patterns. A French reference, for example, can yield intelligible English with a French accent. Accent control must therefore regulate how reference pronunciation influences synthesis while retaining speaker identity.

Accent control can be learned during training or applied during synthesis. Disentanglement methods use multilingual or multi-accented data and task-specific optimization [1, 2, 3, 4]; a TTS→voice-conversion (VC) cascade instead converts every output utterance. For frozen models, Activation Steering [5] uses offline-extracted accent directions, while standard classifier-free guidance (CFG) and its flow-matching adaptations [6, 7, 8] mix conditional and unconditional velocity fields. We use a native speech reference to supply the pronunciation condition and reuse it across utterances, placing accent control inside autoregressive generation.

Our method uses two references with distinct, complementary roles: the original speech anchors speaker identity, while a converted native anchor supplies target-language pronunciation. Both references condition the same generated patch, but mixing their decoder outputs alone leaves open how the two autoregressive states should advance. We feed the emitted patch back to both states, so each branch follows the speech actually produced while retaining its own reference context. The weight λ controls the anchor's contribution, and the same anchor is reused for subsequent utterances.

Contributions:

1. **Closed-loop autoregressive composition.** We address context desynchronization: independently advanced reference branches condition guidance on different generated histories. Feeding each emitted patch back to both LM states preserves a common history while retaining distinct reference contexts. On development, this update reduces WER from 0.139 with independent branch rollouts to 0.004 and raises GOP from 3.524 to 4.791.
2. **Reusable native anchors.** Zero-shot VC constructs a speaker-specific native reference at enrollment, without parallel bilingual recordings or task-specific training. Subsequent utterances reuse this reference, moving conversion out of online synthesis.
3. **Composition in velocity and conditioning space.** The shared state update supports both CFG-Zero* [9] velocity mixing and fusion of decoder conditions. Conditioning fusion uses one conditional branch instead of two, reducing decoder branch evaluations by one third (27 to 18 per patch) while retaining both reference contexts and their feedback.

2. RELATED WORK

Accent conversion and post-hoc steering. Earlier work maps non-native speech toward a native target [10, 11], uses phonetic posteriorgrams [12, 13], or factorizes accent-related attributes in a neural codec [14]. Activation Steering [5] extracts layer-wise accent directions without further training. Our guidance comes from a speaker-specific native anchor and two reference contexts updated with the generated speech. Vevo [15] separates style and timbre through trained modules. We use Seed-VC [16] for both anchor construction and the TTS→VC baseline, comparing a reusable enrollment-time reference with per-utterance conversion.

Cross-lingual TTS and guidance formulations. Zhang et al. [17] studied accent leakage in cross-lingual TTS, followed by training-time disentanglement methods [2, 3, 4, 18]. X-Voice [1] combines pseudo-parallel prompts for training-time adaptation with dropped-condition CFG, $v(\mathrm{ref}) - v(\emptyset)$. CrossAccent-TTS [4] and Accent Vector [18] learn representations or weights for accent control. Our method composes two reference-conditioned states during inference in a frozen TTS model. Dual-scale CFG methods vary guidance between text and speaker conditions [19, 20, 21]; our two contexts each contain a speech reference and follow a common generated history. Section 6.1 compares native-anchor guidance with a reference-free secondary context.

*Corresponding author.

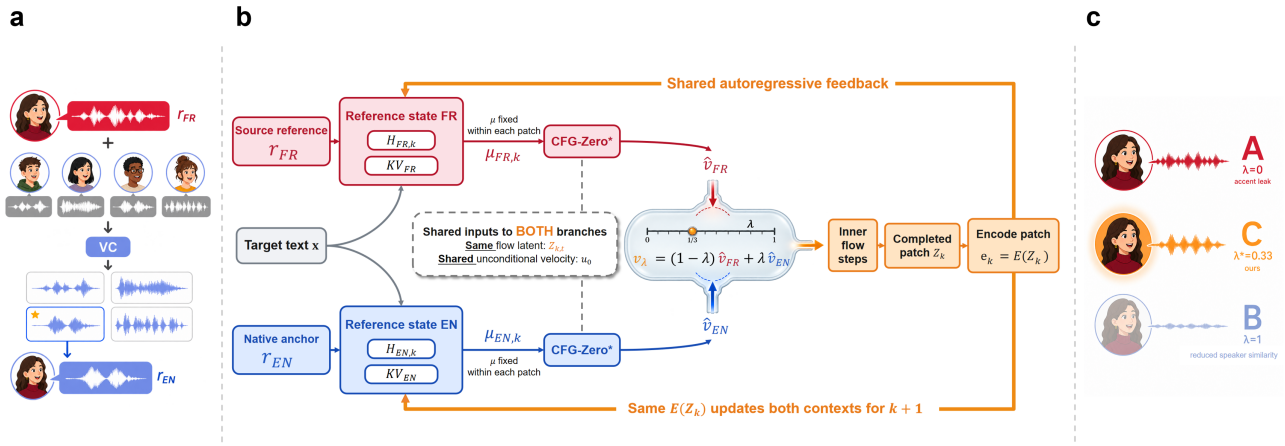

Fig. 1. Method overview. **(a)** Offline native-anchor construction: r_{FR} and four native donors undergo zero-shot conversion once per speaker; the highest-similarity candidate is kept as r_{EN}. **(b)** Closed-loop dual-reference guidance. At autoregressive step k, both reference-conditioned fields are evaluated on the same flow latent $z_{k,t}$ and combined via Eq. (1). After the ODE produces acoustic patch Z_k, its encoding advances both LM contexts via Eq. (2). Shared generated history; distinct reference contexts and KV caches. **(c)** $\lambda = 0$ is Arm A (accent leakage); $\lambda = 1$ is Arm B (reduced speaker similarity); $\lambda^* = 0.33$ is the selected operating point (Arm C).

3. METHOD

3.1. Synthetic native anchor

Let x denote target-language text, and let a frozen zero-shot TTS condition its flow-matching decoder on reference speech r, yielding $v(z, t; x, r)$. For each speaker we synthesize a native-English anchor r_{EN} with a zero-shot voice converter (Seed-VC) [16], converting a native English donor toward the timbre of the source reference r_{FR}.

Donors are a curated LibriTTS-R [27] subset (highest-DNSMOS utterance from each of the top 20 native English speakers). With a fixed seed, four distinct donors are cyclically assigned per speaker. Seed-VC with length adjustment 1.0 preserves donor duration (3.04–9.92 s). We keep the candidate maximizing $\mathrm{SIM}_{\mathrm{avg}} = (\mathrm{SIM}_{\mathrm{ECAPA}} + \mathrm{SIM}_{\mathrm{CAM++\text{-}CN}})/2$ against r_{FR}. The four conversions are performed at enrollment; the selected anchor is reused for subsequent utterances (Fig. 1).

3.2. Closed-loop dual-reference guidance

Generation has an outer autoregressive patch index k and an inner flow time t. Let $H_{X,k}$ denote the base- and residual-LM state, including their KV caches, for reference context $X \in \{\mathrm{FR}, \mathrm{EN}\}$. Each context is initialized by jointly prefilling the target text and local-encoder features of r_X. At step k, the LMs provide diffusion-transformer (DiT) conditioning $\mu_{X,k}$, which remains fixed during that patch's ODE solve. Both fields are evaluated on the same evolving latent $z_{k,t}$ and combined with CFG-Zero* [9]. Suppressing k for clarity, define $u_X = v(z, t; x, \mu_X)$ and the shared unconditional velocity $u_0 = v(z, t; x, \emptyset)$. With stabilizer $\epsilon = 10^{-8}$ and guidance scale $s = 2.0$,

$$\begin{cases} \alpha_X = \dfrac{\langle u_X, u_0 \rangle}{\|u_0\|_2^2 + \epsilon}, \\ \widehat{v}_X = \alpha_X u_0 + s\big(u_X - \alpha_X u_0\big), \\ v_\lambda = (1-\lambda)\,\widehat{v}_{\mathrm{FR}} + \lambda\,\widehat{v}_{\mathrm{EN}}. \end{cases} \tag{1}$$

CFG-Zero* splits the conditional field into components parallel and orthogonal to u_0 via α_X. The first Euler update is zeroed (fraction 0.04); the remaining nine follow Eq. (1). Sampling is fixed (sway 1.0, temperature 1.0, 10 Euler steps).

After the ODE produces acoustic patch Z_k, its local encoding and projection $E(Z_k)$ advance both LM contexts:

$$H_{X,k+1} = F\big(H_{X,k}, E(Z_k)\big), \qquad X \in \{\mathrm{FR}, \mathrm{EN}\}. \tag{2}$$

Here F is the frozen base- and residual-LM update. Both contexts follow the emitted history while retaining distinct reference prefills. The open-KV ablation uses independently sampled histories (Sec. 6.1).

The same update supports a conditioning-space implementation: $\mu \leftarrow (1-\lambda)\mu_{\mathrm{FR}} + \lambda\mu_{\mathrm{EN}}$. This variant retains both contexts and Eq. (2), then uses a single conditional DiT branch plus the unconditional branch (Secs. 5.3, 6.1). Velocity mixing is used for all held-out test and transfer runs. $\lambda = 0$ gives A, $\lambda = 1$ gives B, and intermediate values give C. We swept $\lambda \in \{0.1, 0.2, 0.25, 0.33, 0.5, 0.75\}$ on development data and fixed $\lambda^* = 0.33$ before evaluating the test set; throughout, λ^* denotes this selected operating point rather than an optimized value.

4. EXPERIMENTAL SETUP

Models. Frozen VoxCPM2 [28] (2B, 48 kHz, tokenizer-free, no LoRA). Anchors and the cascade use Seed-VC [16] (Whisper-small + BigVGAN-v2; 25 steps, CFG 0.7, length adjustment 1.0).

Arms. We refer to each compared configuration as an arm. A: real r_{FR}. B: synthetic r_{EN}. C: proposed guidance at $\lambda^* = 0.33$. Cascade: unconditioned English from VoxCPM2, then Seed-VC transfers r_{FR} timbre on every utterance.

Data. Development (VP-Stress): 30 VoxPopuli [29] French speakers (LID-FR $\geq$ 0.9, reference 3–10 s), 10 English texts, seed 20260822 (n = 300/arm). Held-out test (B1-CNS-test): 20 MLS [30] French speakers $\times$ 20 English texts $\times$ 3 seeds, item-matched $n = 1200$/arm including the cascade; speakers are disjoint from development, donors, and tuning. All 20 test anchors pass $\mathrm{SIM}_{\mathrm{avg}} > 0.45$ (mean 0.707). Generalization: 15 VoxPopuli Spanish speakers, 10 English texts, with $\lambda^* = 0.33$ transferred without re-tuning.

Table 1. Main evaluation on held-out test data (20 French speakers × 20 texts × 3 seeds; $n = 1200$ per arm, $\lambda^* = 0.33$). Speaker-macro averages; bold values mark the best result in each column. $P(\mathrm{fr})$ is residual French LID mass. *CAM++-CN is Seed-VC's internal speaker encoder (Sec. 6.2).

Arm	WER ↓	$P(\mathrm{fr})$ ↓	$P(\mathrm{en})$ ↑	Speaker similarity ↑			GOP ↑	DNSMOS ↑
				ECAPA	CAM++-EN	CAM++-CN*		
A (Direct)	0.016	0.093	0.843	**0.601**	**0.624**	0.746	4.663	2.900
B (Anchor)	0.013	**0.000**	**0.993**	0.404	0.444	0.702	4.733	2.934
C (Proposed)	**0.009**	0.025	0.942	0.559	0.586	0.748	**4.750**	2.887
Cascade	0.047	0.001	0.985	0.468	0.492	**0.770**	4.378	**3.026**

Metrics. WER and marginal $P(\mathrm{fr})$ / $P(\mathrm{es})$: faster-whisper-large-v3 [31] (beam 1, no VAD; full softmax, not top-1). GOP [32]: wav2vec2-xls-r-300m-phoneme [33] posteriors, g2p_en, CTC alignment; utterance GOP averages $\log P(y) - \max_{p \neq y} \log P(p)$ over target phones y. Similarity: cosine vs. the foreign reference on ECAPA-TDNN [34], CAM++-EN, and CAM++-CN [35]. We report all three encoders and listener preferences to cover language/accent sensitivity [22, 23, 24] and differences between automatic and perceptual similarity judgments [25, 26]. Quality: DNSMOS P.835 OVRL [36]. Statistics: speaker-macro means, paired deltas, 10,000 speaker-clustered bootstrap 95% CIs; preferences use session majority and a speaker-clustered two-sided exact sign-flip test.

5. RESULTS

5.1. Development trade-off and λ^* selection

At $\lambda = 0.33$, the GOP gain over A is $+0.189$, nearly matching the $+0.193$ attained at $\lambda = 0.5$, while the ECAPA similarity cost is roughly halved ($\Delta\mathrm{ECAPA} = -0.049$ $[-0.063, -0.036]$; Fig. 2). CAM++-EN follows the same trend. At this setting, paired differences are $+0.021$ $[-0.062, 0.098]$ in GOP relative to B and -0.018 $[-0.068, 0.032]$ in DNSMOS relative to the cascade. We selected $\lambda^* = 0.33$ on development and fixed it for test and Spanish transfer.

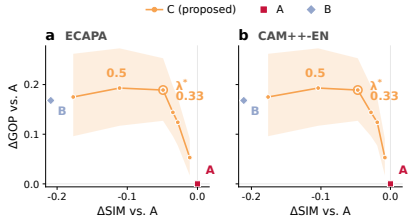

Fig. 2. Accent–timbre trade-off on development (30 speakers). **(a)** ECAPA; **(b)** CAM++-EN. x: ΔSIM vs. A; y: ΔGOP vs. A. The shaded band is the 95% speaker-clustered bootstrap CI of ΔGOP. Orange: C (proposed) λ sweep; λ^* is the selected operating point (0.33).

5.2. Held-out test evaluation (λ^* frozen)

C vs. A. At the fixed operating point, C achieves mean WER 0.009 and GOP 4.750, the lowest and highest values, respectively, among the held-out test arms (Table 1). Relative to direct cloning, it reduces residual French LID mass by 73% ($0.093 \rightarrow 0.025$; $\Delta = -0.068$ $[-0.119, -0.027]$) and increases GOP by 0.087 $[0.048, 0.128]$ (Fig. 3). The paired WER difference is -0.0072 $[-0.0147, 0.0003]$, with an ECAPA change of -0.042 $[-0.048, -0.036]$.

C vs. B. Adding the original-reference branch recovers 0.155 ECAPA similarity ($[0.137, 0.173]$) and 0.142 CAM++-EN similarity ($[0.122, 0.162]$) over anchor-only cloning. This speaker-similarity recovery comes with higher mean GOP (4.750 vs. 4.733) and lower mean WER (0.009 vs. 0.013); paired differences are $+0.017$ $[-0.043, 0.072]$ and -0.004 $[-0.011, 0.002]$.

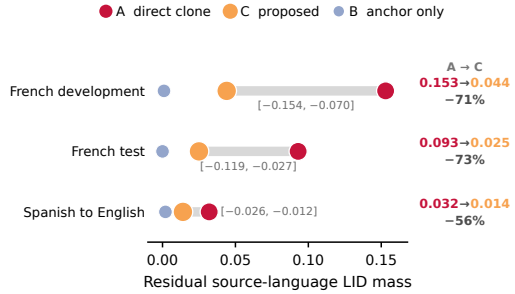

Fig. 3. Residual source-language LID mass at $\lambda^* = 0.33$. Dumbbells: A→C reduction; B is the floor. Brackets give speaker-clustered 95% CIs of the paired difference $\Delta = C - A$ (negative values indicate a reduction under C).

5.3. Comparison with the TTS→VC cascade

On the item-matched three-seed test set ($n = 1200$/arm), C reduces mean WER from the cascade's 0.047 to 0.009, an approximately 81% reduction ($\Delta = -0.038$ $[-0.046, -0.031]$), and raises GOP from 4.378 to 4.750 ($\Delta = +0.372$ $[0.330, 0.412]$). Word repetition or insertion involving at least two words occurs in 3.2% (38/1200) of cascade outputs, versus less than 0.5% for A, B, and C.

Speaker similarity also improves over the cascade: ECAPA increases by 0.091 (95% CI $[0.073, 0.109]$) and CAM++-EN by 0.094 ($[0.073, 0.114]$). In the blinded development study, listeners likewise prefer C on speaker identity (38 vs. 5 sessions, $p < 0.0001$; Sec. 5.5). On the remaining metrics, C trails the cascade: C−cascade is $+0.024$ on residual $P(\mathrm{fr})$ ($[0.005, 0.052]$; cascade mean 0.001), -0.139 on DNSMOS ($[-0.197, -0.089]$), and -0.022 on CAM++-CN (CI $[-0.039, -0.006]$; Sec. 6.2). Thus, the cascade's higher DNSMOS accompanies more word errors, while C improves both transcription accuracy and listener-rated speaker identity.

Table 2. H100 80GB inference. **(a)** vs. direct cloning and cascade. **(b)** Matched velocity vs. conditioning (μ-mix). TTS: 9-word text, five timed runs after warmup. DiT: batched branch evaluations per patch over nine nonzero Euler updates. †sequential cascade stages.

Arm	DiT/patch	VC NFE	Lat. (s)	RTF	Peak VRAM (MiB)
(a) Direct cloning and cascade comparison					
A	18	—	1.27	0.474	8239
C (velocity)	27	—	1.52	0.587	8235
Cascade	18	50	≈2.10	≈0.80	8563 / 1474†
(b) Matched implementation comparison					
C (velocity)	27	—	1.62	0.589	8406
C (conditioning)	18	—	1.47	0.572	8408

Compute. C (velocity) achieves approximately 26% lower online RTF than the cascade (0.59 vs. 0.80). Relative to direct cloning, it adds 0.25 s with essentially unchanged peak memory. Four an-

chor conversions cost approximately 3.1 s per speaker. At the benchmarked length, $T_C(N) \approx 3.1 + 1.52N$ and $T_{\text{cascade}}(N) \approx 2.10N$; the reusable anchor amortizes conversion cost by six utterances.

Conditioning-space implementation. C (conditioning) reduces DiT branch evaluations from 27 to 18 per patch and retains both LM contexts. In the matched rerun, its RTF is 0.572 versus 0.589 for C (velocity), with nearly identical peak memory. RTF reductions across 5-, 9-, and 20-word texts are 0.7–2.9%.

5.4. Cross-language generalization (Spanish→English)

The French-selected $\lambda^* = 0.33$ transfers directly to 15 Spanish speakers without language-specific tuning. Residual Spanish LID mass drops from 0.032 to 0.014 ($\Delta = -0.018$ $[-0.026, -0.012]$; Fig. 3), and ECAPA similarity recovers from 0.323 to 0.474 relative to B. C achieves WER 0.0055, compared with 0.0054 for A ($\Delta = +0.0001$ $[-0.0070, 0.0065]$). The same operating point therefore combines source-LID reduction with speaker-similarity recovery in both French-to-English and Spanish-to-English cloning.

5.5. Subjective human evaluation

A randomized, blinded preference study used seven native-English evaluators and 60 sessions (30 development speakers × 2 texts), with the French reference and outputs from A, B, C ($\lambda^* = 0.33$), and the cascade. Evaluators chose the baseline, C, or a tie. Four pre-planned comparisons covered accent (A vs. C, C vs. cascade) and speaker similarity (B vs. C, C vs. cascade). A follow-up comparison assessed C vs. A speaker similarity.

Table 3. Preferences on development speakers (7 evaluators × 60 sessions). % (sessions), majority vote. Bold: significantly preferred system. p: speaker-clustered sign-flip; pre-planned rows remain significant under Holm–Bonferroni. †Follow-up speaker-similarity comparison.

Comp.	Dim.	Pref. base	Pref. C	Tie	p
C vs. A	Accent / nat.	6.7 (4)	**28.3 (17)**	65.0 (39)	0.0057
C vs. B	Speaker sim.	5.0 (3)	**65.0 (39)**	30.0 (18)	< 0.0001
C vs. Casc.	Accent / nat.	**53.3 (32)**	21.7 (13)	25.0 (15)	0.026
C vs. Casc.	Speaker sim.	8.3 (5)	**63.3 (38)**	28.3 (17)	< 0.0001
C vs. A†	Speaker sim.	26.7 (16)	21.7 (13)	51.7 (31)	0.723

Listeners significantly prefer C to direct cloning on pronunciation naturalness (17 vs. 4 sessions, $p = 0.0057$; Table 3). In the follow-up speaker-similarity comparison with direct cloning, 51.7% of sessions are ties; the remaining sessions split 13 for C and 16 for A ($p = 0.723$). Relative to anchor-only cloning, C wins 39 vs. 3 sessions on speaker identity ($p < 0.0001$). The comparison with the cascade separates pronunciation from identity: listeners favor the cascade on pronunciation naturalness (32 vs. 13, $p = 0.026$), and C on speaker identity (38 vs. 5, $p < 0.0001$). Fleiss' κ is 0.167–0.256; 87–89% of majority-disagreement judgments are ties.

6. ANALYSIS

6.1. Guidance mechanisms and architectural ablations

Shared autoregressive feedback. The open-KV ablation tests the effect of independently advancing the reference branches. Both configurations mix velocities for the emitted patch, but open-KV feeds each LM its own separately sampled, unmixed prediction. This permits context desynchronization: each branch follows its own generated history rather than the emitted sequence. Replacing these independent rollouts with shared patch feedback reduces WER from 0.139 to 0.004 and raises GOP from 3.524 to 4.791. Equation (2) enforces the common-history constraint at every patch while preserving each branch's reference context.

Reference contexts. C (velocity) gives ECAPA 0.467 versus 0.372 for concat; concat reaches $P(\text{fr}) = 0.003$. Replacing the native anchor with $\emptyset$ lowers similarity and GOP. The waveforms used by wavavg differ in length and phase; their average gives GOP 3.131.

Composition across mixing sites. C (conditioning) achieves ECAPA 0.465 and GOP 4.828, versus 0.467 and 4.791 for C (velocity). The paired differences are not significant (velocity minus conditioning: ECAPA $+0.003$ $[-0.004, 0.009]$; GOP -0.038 $[-0.089, 0.014]$). Both implementations retain separate reference states and common patch feedback; conditioning mixing achieves RTF 0.572 (Table 2).

Table 4. Paired ablations on development (30 speakers, $\lambda = 0.33$, $n = 300$).

Config.	ECAPA ↑	CAM++-EN ↑	GOP ↑	WER ↓	$P(\text{fr})$ ↓
A ($\lambda = 0$)	0.517	0.589	4.602	0.020	0.153
B (anchor only)	0.308	0.378	4.770	0.009	0.001
C (velocity)	0.467	0.542	4.791	0.004	0.044
C (conditioning)	0.465	0.546	4.828	0.000	0.034
concat	0.372	0.459	4.743	0.003	0.003
open-KV	0.458	0.522	3.524	0.139	0.045
C_{noref} ($\emptyset$)	0.441	0.517	4.653	0.010	0.063
wavavg	0.359	0.420	3.131	0.011	0.008

6.2. Encoder-domain discrepancy in speaker verification

The three encoders assign substantially different similarity changes to the same accent-guided speech. On French development, C vs. A at $\lambda = 0.5$ is -0.111 on ECAPA and -0.104 on CAM++-EN, but -0.020 on CAM++-CN. The CAM++ checkpoints share an architecture but were trained on VoxCeleb2 and a 200k-speaker Chinese dataset, respectively; their responses show that the estimated identity cost of accent control depends on the verification model. At $\lambda^* = 0.33$, the CAM++-CN difference is $+0.003$ $[-0.003, 0.008]$. CAM++-CN also serves as Seed-VC's internal speaker encoder and contributes to our anchor-selection score (Sec. 3.1). On the held-out test, ECAPA and CAM++-EN rank C above the cascade, whereas CAM++-CN reverses that ordering. The blinded listening study independently favors C on speaker identity (38 vs. 5 sessions; Sec. 5.5), aligning with the ECAPA and CAM++-EN ranking.

7. CONCLUSION AND LIMITATIONS

We presented closed-loop dual-reference guidance for accent control in frozen TTS. Separate reference states combine original-speaker information with a reusable native anchor, and feedback from the emitted patches keeps both branches on the same generated history; this common-history constraint, rather than the choice of mixing site, accounts for the largest quality gain (Sec. 6.1). On held-out French speakers, the method cuts residual French LID mass by 73% versus direct cloning while recovering 0.155 ECAPA similarity over anchor-only cloning, and the French-selected operating point transfers to Spanish without re-tuning. Native-English listeners prefer C on pronunciation naturalness over A and on speaker identity over B and the cascade. The current evaluation targets English only, anchor quality depends on the zero-shot converter, and the guidance weight is fixed across patches; extending the evaluation to additional target languages, adapting guidance per patch, and improving converted-anchor quality are the next steps.

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